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Industrial Hygiene Recordkeeping

1.0 GENERAL

All exposure assessments that qualitatively or quantitatively estimate employee exposures to chemical or physical agents should be retained indefinitely. This practice provides historical documentation in the event an employee or ex-employee develops a chronic adverse health effect that may be related to his or her occupational exposure. It also provides the epidemiologist with useful information in the event of a retrospective health study such as the recently performed study by the University of California at Davis sponsored by the Semiconductor Industry Association.

U.S. regulations require that employee exposure records be maintained for at least thirty years.[1] Exposure records for certain chemicals (e.g., arsenic, lead, etc.) must be retained for at least forty years, or for the duration of employment plus twenty years, whichever is longer.[2] At the expiration of the retention period, advanced notice must be given to OSHA before disposal of the records. Records of employee exposure to ionizing radiation required by U.S. regulations must be retained indefinitely.(3)

Exposure records that are retained should include:

1. Name of individual, equipment, or area assessed
2. Identifying numbers (e.g., employee number, equipment model number, social security number, etc.)

3. Operation performed during the assessment and a brief description of the process, including the chemical or physical agents of concern and potential sampling/analysis interferences
4. Calibration data for the sampling equipment and analytical technique, if applicable
5. Type of sample if applicable (i.e., personal, breathing zone, or area sample)
6. Date and time the samples were taken
7. Analytical procedure if applicable (e.g., NIOSH or OSHA method)
8. Sample and analytical method if applicable
9. Analytical results if applicable, including any calculations
10. Additional observations which may help to characterize the survey

When a personal sample is taken, a written copy of the sample results should be included in the person's medical file. This strategy allows review by the physician or occupational health nurse. This allows the occupational health provider to have a greater awareness of types of chemicals and physical agents the employee works with and their exposure levels.

Under U.S. regulations, if a company goes out of business, medical and exposure records are required to be transferred to the successor employer. If there is no successor employer, the records are sent to the appropriate government agency.[4]

2.0 CONTINUOUS MONITOR RECORDS

Within the semiconductor industry, a variety of continuous gas monitors are used (see Ch. 2, Sec. 3.2: Continuous Monitoring, for additional information). Historically, these monitors were primarily used to determine if a toxic gas leak had occurred, rather than actual measurements of breathing zone concentration of chemicals. Sample points for the monitors were located in exhausted enclosures as well as at potential release points in the clean room.

Many of the monitors—particularly older models—gave their results in the form of a hardcopy readout. As large quantities of the readouts began to pile up, semiconductor companies were faced with the issue of what to retain and what to discard. Different approaches to this issue include:

1. Retain all hardcopy readouts—with the uncertainty of being able to retrieve data on a specific event
2. Microfiche hardcopy readouts
3. Retain all read-outs documenting any detectable leakage or calibration issues
4. Retain only those read-outs documenting a release that resulted in an alarm level or are required to be retained by regulation, such as Fed OSHA's 29 CFR 1910.20 which requires that all employee exposure records (specifically "environmental [workplace] monitoring or measuring of a toxic substance or harmful physical agent, including personal, area, grab, wipe, or other forms of sampling") be retained[1]

3.0 VENTILATION RECORDS

In the semiconductor industry, local exhaust ventilation is a key control for the prevention of employee exposure to volatile chemicals (see Ch. 4: Ventilation). As such, exhaust ventilation measurement records are retained to allow the evaluation for trends in ventilation system performance and to provide exposure control information in the event of possible litigation.

4.0 EMPLOYEE COMMUNICATION

The semiconductor industry has typically gone beyond what is required by government regulations for communicating exposure results to employees. Within the semiconductor industry, the results of exposure assessments are typically communicated to all affected employees, not just those who participated in the survey and/or those who are required to be notified by regulation. Methods of communication commonly used include posting on bulletin boards, written memos summarizing the results, and presentations at informal management/employee meetings. The suitability of the particular mechanism used to accomplish this varies according to a number of factors such as the concern level of the employees, complexity of the survey, the exposure levels found, and the methods used by the site to communicate other employee related information (such as benefits modifications, company performance, policy changes, etc.).

5.0 PERSONNEL RECORDS

The incorporation of IH records into the personnel files of employees is a practice that is very much open to conjecture. Typically, this practice is used where there is not an on-site medical department that can act as a defined repository for important (and legally required) IH or medical records to be retained. In this case, and with the concurrence of the companies' legal and human resources departments, the retention of this category of records may be the only viable alternative available.

REFERENCES

1. U.S. Code of Federal Regulations, 29 CFR Part 1910.20 (d) (1)(ii), U.S. Government Printing Office, Philadelphia, PA
2. U.S. Code of Federal Regulations, 29 CFR Part 1910.1018 (q) (1)(iii), U.S. Government Printing Office, Philadelphia, PA
3. U.S. Code of Federal Regulations, 29 CFR Part 1910.96 (n) (1), U.S. Government Printing Office, Philadelphia, PA
4. U.S. Code of Federal Regulations, 29 CFR Part 1910.20 (h), U.S. Government Printing Office, Philadelphia, PA